NAME : N.S.MUNITEJA

REG NO : 192312388

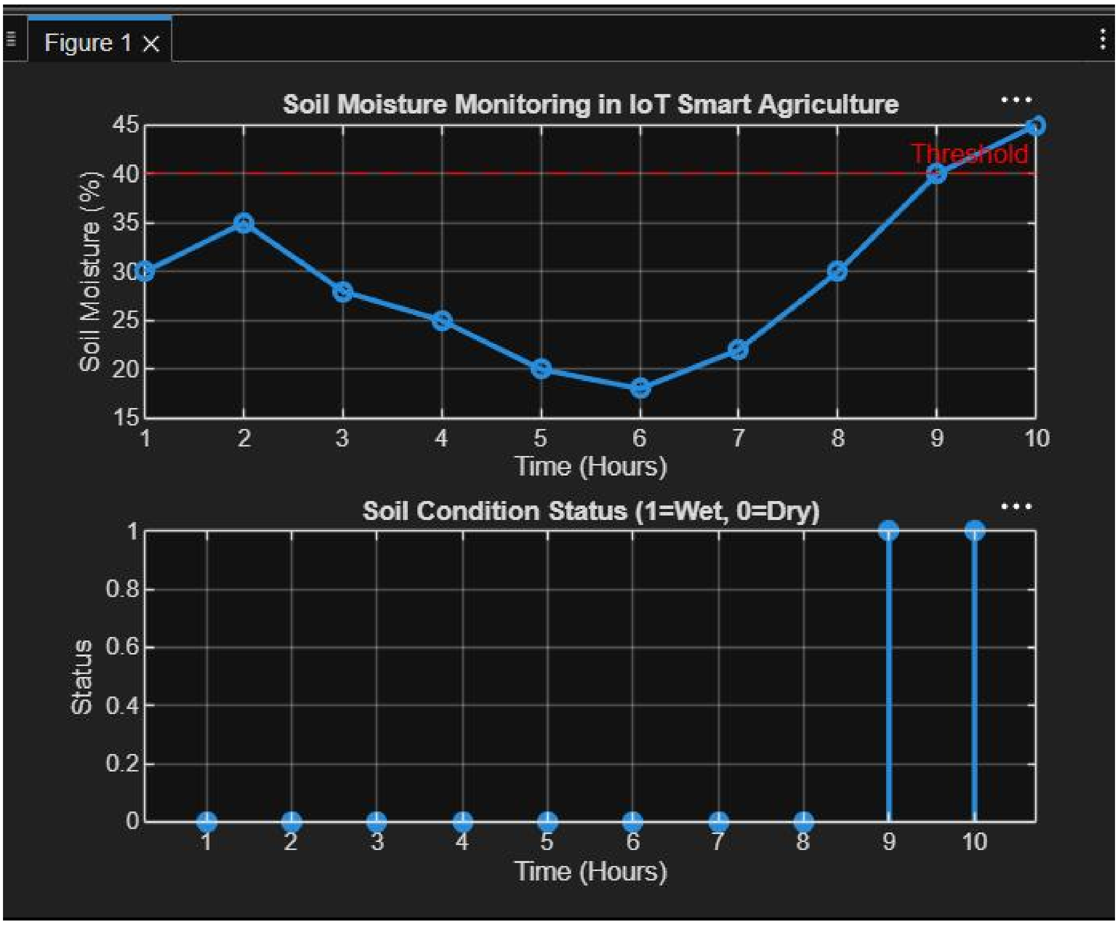
Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

OBJECTIVE 1:

CODE:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Moisture Status (Dry/Wet Level) --------
```

```
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
```



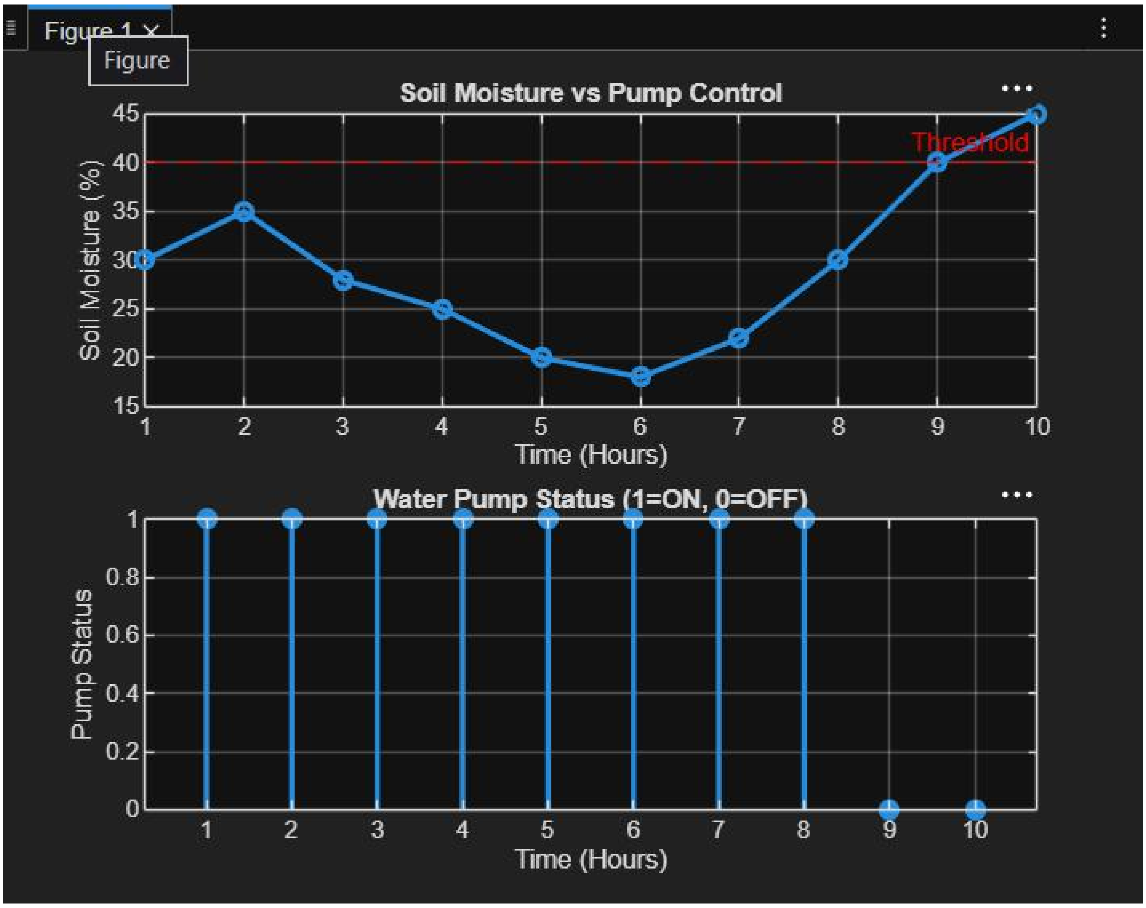
OBJECTIVE 2:

CODE:

```
clc; clear; close all;
% Simulated soil moisture (%)
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')

% -------- Graph 2: Pump Status (ON/OFF) --------
subplot(2,1,2)
stem(time, pump_status, 'filled','LineWidth',2)
title('Water Pump Status (1=ON, 0=OFF)')
xlabel('Time (Hours)')
ylabel('Pump Status')
grid on
```



OBJECTIVE 3:

CODE:

```
clc;
clear;
close all;

% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level
```

```
threshold = 40;

% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % Limit to 100%

figure

% -------- Graph 1: Soil Moisture vs Time --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold, 'r--', 'Threshold')

% -------- Graph 2: Irrigation Efficiency --------
subplot(2,1,2)
plot(time, efficiency, '-s', 'LineWidth', 2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
```

grid on

ylim([0 100])